

Stereotaxic surgery protocol (by PA, Jul 2020; based on GP and AH protocols + tips from JS, GL and TL)

Our surgery room: Ward 15-106. Reserve surgery time in our google calendar.

Prior to surgery – in lab

- Get ice to transport virus aliquots.
- Check available virus aliquots. If virus is not aliquoted, go to **Appendix 1 – Aliquoting viruses**.
- Dilute viruses (if needed) using sterile saline – we only have a 10 ul pipette in the surgery suite.
- Check available buprenorphine (0.02 mL per mouse) in histology room (pump tools drawer).



- If buprenorphine syringe has yellow needle (large gauge, used to collect buprenorphine from its vial faster), swap for brown one (smaller gauge, used for subcutaneous injection). We use undiluted buprenorphine (1 mg/ml).
- Write down mouse cages you will use.
- Grab personal supplies (wildcard, phone, surgery sheet, DIY opto probes).
- Grab any re-stocking supplies (if needed).

Prior to surgery – in surgery suite

- Turn on germinator (bead sterilizer) – it takes about ~1h to warm up. When ready, green LED will turn ON.
- Check levels of iso in anesthesia machine. If not full, go to **Appendix 2 – Filling iso tank/vaporizer**.
- Weigh activated charcoal canister (isoflurane collection cylinder).
- Check available oxygen gas. If pressure is low, go to **Appendix 3 – Removing and installing gas regulator**.
- Check that stereotax arms are firmly secured (not wobbly).
- Prepare a syringe with carprofen 1 mg/ml.
 - If diluted carprofen is available (fridge), shake bottle before collection.
 - To dilute from 50 mg/ml stock (fridge): collect 0.2 ml from stock and inject it into a 10 ml sterile saline vial. Label.

Gather on top of a sterile pad or near your station:

- Syringe with buprenorphine 1 mg/ml (0.02 mL per mouse) – opioid anesthetic
- Syringe with carprofen 1 mg/ml (0.1 ml per mouse) – NSAID
- Shot glass of Nair (hair/fur removal)
- Shot glass of Betadine (iodine – antiseptic)
- Cotton swabs
- Triangle sponges
- Alcohol wipes
- Scalpel blade
- Parafilm pieces
- Vetbond (glue)
- Eye ointment
- Ear tags + tagger
- Sutures (not needed if doing implants)
- Plastic beaker + black bag (for waste)

Gather metal tools on top of your sterile pad after sterilizing them in germinator for ~20 s:

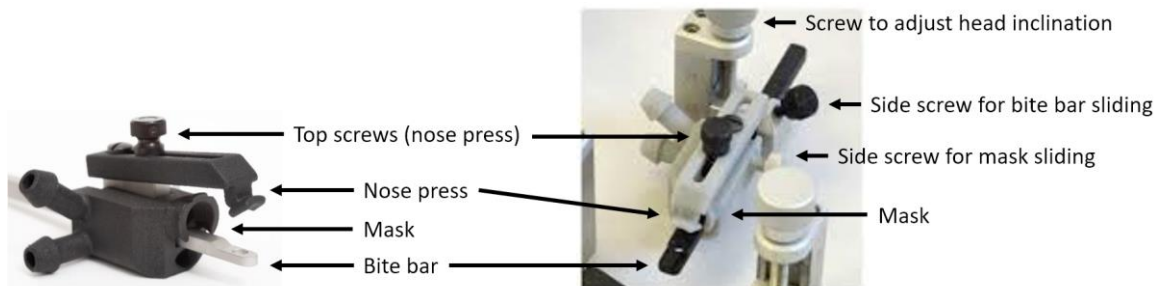
- Skull scraper
- Scalpel
- Forceps
- Hemostat with scissors (2 if doing sutures; 1 if doing implants)
- Small scissors (if doing implants)

If doing implants, also gather:

- Cardboard enrichment (mouse room) to swap for plastic enrichment in cages.
- Porcelain dish for mixing metabond
- Metabond powder (green container) + white scoop
- Metabond liquid (blue cap)
- Metabond catalyst (golden syringe)
- Green metabond applicators
- Flow-It ALS Light-Curing cement

Turn ON:

- Lights
 - Drill
 - Digital stereotaxic display
 - Syringe pump controller
 - Heating pad – plug battery in and use at setting 4
 - Recovery heating pad – medium
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- Clean NanoFil syringe(s) with water and then saline (use fresh solutions in a 24-well plate).
 - Insert NanoFil syringe(s) into pump(s). Pull the plunger a little bit to fit.
 - Carefully maneuver syringes to avoid bending the delicate needle. If you damage/bend the needle, replace it.
 - Note that brand new syringes are shipped with a large gauge needle. Replace the needle before using.
 - Loosen screws for nose press and iso mask sliding. Move them back to expose bite hole in bite bar.



Surgery I – Anesthesia and stereotax

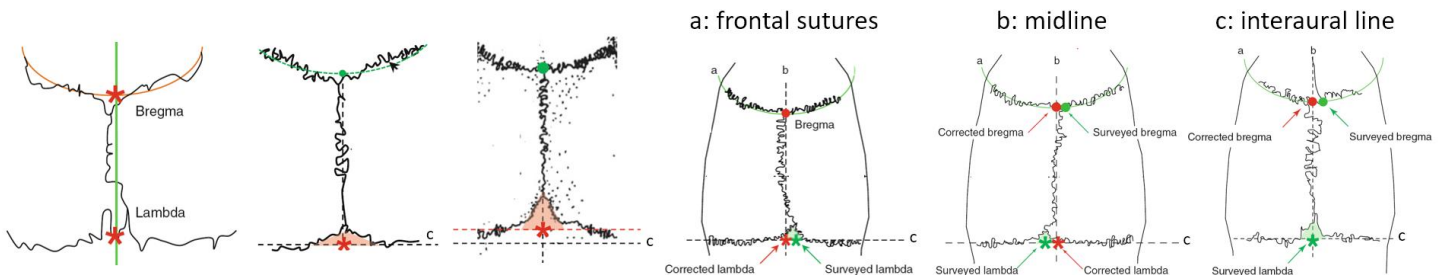
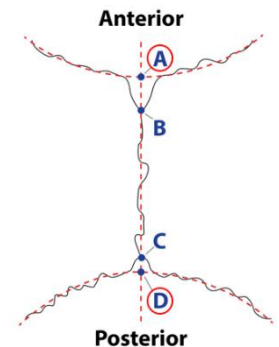
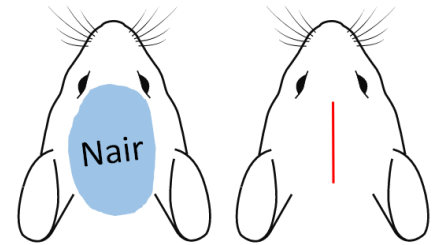
- Open oxygen gas tank.
- Turn ON anesthesia machine – ON/OFF switch & system pressure switch.



- Open iso vaporizer on anesthesia machine – set it to 4.
- Turn ON flow switches for induction chamber (follow color coded tubing).
 - If flow indicators do not float, something is wrong – check that oxygen tank is open/has gas.
- While induction chamber fills up with iso, grab cage from mouse room.
- Put mouse inside induction chamber for 30 s to 1 min and monitor breathing.
 - Be patient – insufficient time in induction chamber will require higher levels of iso during surgery.
- After mouse breathing slowed significantly (~1 breath/s), turn OFF flow switches for induction chamber.
- Turn ON flow switches for iso mask.
- Drop iso level from 4 to 3.
- Remove mouse from induction chamber and put it on the stereotax. Heating pad should be warm.
- Scruff mouse and pry its mouth open with the back of a cotton swab. Place mouse's top teeth on bite bar.
 - Be gentle to not wound the mouse's tongue and mouth.
- Move iso mask over the mouse's nose. If mask does not move, loosen its side screw.
 - Be careful not to suffocate mouse – if you notice strong gasping, move mask a bit.
- Adjust mouse's limbs to a more comfortable position.
- Cover eyes with eye ointment to prevent them from drying out.
- Inject 0.02 ml of buprenorphine subcutaneously.
 - Inject slowly and wait a few seconds before removing needle because drug is viscous.
- Inject 0.1 ml of carprofen subcutaneously.
- Adjust ear bars to secure skull.
 - Grooves are below/in front of the mouse's ears. We do NOT place them inside the ear canal.
 - You might have to adjust height of ear bars (keep them symmetrical), position of bite bar and/or head inclination.
 - Try to move head side-to-side with your fingers – it should not move.
 - You might hear a soft click when tightening the ear bars – that is normal.
- Adjust nose press (optional).
- From now on, monitor mouse's breathing – if too fast, increase iso flow rate; if too slow, decrease iso flow rate.
- Monitor mouse's pain – mouse should NOT react to toe pinches; if it does, increase iso flow rate.
- Monitor heating pad battery and swap battery packs when battery is low.
- As the surgery progresses, you may drop the iso level to 1-2%.

Surgery II – Drilling holes in skull

- With a cotton swab, apply Nair from behind the ears to just behind the eyes.
 - Going against the grain helps.
- Let Nair sit for 15-30 seconds or keep on spreading it until fur comes off.
- Rotate the cotton swab against the grain to collect as much fur as possible.
- Use a new cotton swab to clean any Nair that got too close to the eyes.
- Use an alcohol wipe to remove all Nair and to remove fur from the area.
- Apply betadine to the exposed skin with a soaked cotton swab.
- Use the swab to move remaining fur away from the area you will operate on.
- Use an alcohol wipe to remove the betadine and any leftover fur.
 - Removing loose fur from the area avoids accidentally leaving fur inside the incision later.
- Do a toe squeeze and a toe pinch to check that mouse is fully anesthetized before moving on.
- Use scalpel to make a midline incision (~1-2 cm long) from behind the eyes to the ears.
- If doing implants, make a larger incision or use hemostat and scissors to remove a bit of skin.
- Use skull scrapper and triangle sponges (or cotton swabs) to clean tissue from skull.
- Scrape until bregma and lambda are clear.
- If doing implants, use scrapper to score skull.
- Remove debris with triangle sponges and/or cotton swabs.
- Attach drill to stereotactic arm and zero the drill at bregma.
 - Bregma and Lambda ARE NOT at the intersection of bone sutures (B and C).
 - They are at the projected intersection of the sutures on top of the midline (A and D).



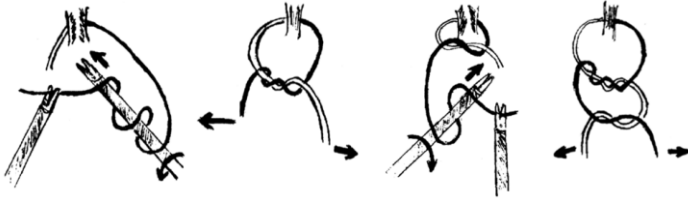
Source of images: Ferry et al 2012 Springer (<https://doi.org/10.1007/978-2-8178-0472-9>).

- Beware that some labs/papers might use different bregma and lambda landmarks and their stereotactic coordinates will need adjustment and in-house histology confirmation.
- For DV bregma zero-ing, drill should touch skull but not deform/put pressure on it.
- Do not confuse bregma for the small suture directly behind the eyes. Bregma is posterior to the eyes by a few mm.
- Move drill to lambda and check if skull is flat/level. In the z-plane (DV), lambda should be at ± 0.1 from bregma. Adjust head inclination if skull is not level.
- Re-zero the drill at bregma and move it to desired coordinates. Do NOT rely on the $\pm$ values of the digital stereotaxic display. Instead, pay attention to where you are moving (dorsal x ventral; medial x lateral; anterior x posterior).
- Use foot pedal to operate drill; start/stop drill away from skull to reach desired speed.
- Move skin away from drill with scrapper if needed.
- Drill through skull until ball tip of drill is through. Check if cortex is visible. Avoid drilling too deep and damaging cortex.
- If there is bleeding, absorb it with triangle sponges. If doing implants, be extra careful to remove debris and blood clots from holes – applying a small drop of saline with the scrapper & cleaning skull with sponge/swab can help.
- Drill all holes.
- If doing implants, further score skull with scalpel. Dry the skull with ethanol wipe and sponge/swabs.

Surgery III – Injections

- Replace the drill on the stereotactic arm with your syringe holder.
- Plan your movements to avoid damaging needle.
- Adjust stereotax arm all the way up and away from ear bars to avoid hitting needle.
- Double check the configuration of injection volume and speed.
- Press and hold the big button that shows up when you turn the syringe controller ON.
- Select DISPLAY ALL CHANNELS.
- If the controller complains about re-setting positions: select pump being used > CONFIGURE > RESET POS > END STOP > press and hold SET STOP until you hear a buzz. Go back to main menu.
- CONFIGURE > change volume target and delivery rate. Go back to main menu.
- Press and hold the button for pump being used > CONTINUOUS INFUSE until plunger is all the way in. Later, press and hold MOMENTARY WITHDRAW (fast) and MOMENTARY INFUSE (fast) to fill and test syringe.
- Place a small piece of Parafilm on the mouse's head.
- Quickly spin the virus (< 2 s) with the bench centrifuge and pipet it onto the piece of Parafilm.
 - Pipet ~double the volume of virus you need to account for losses along the way.
- Aspirate the virus into the needle using the syringe controller (momentary withdraw).
 - If there are bubbles in your syringe, eject all reagent and aspirate again.
- Discard parafilm (your black waste bag will go into the biohazard waste).
- Zero the needle at bregma and move it to the desired coordinates (needle and drill will zero at different positions).
- Eject a bit of virus with the syringe controller (momentary infuse) to check that your setup is working well.
- Absorb little bubble of virus with a triangle sponge.
- Lower needle into brain slowly (min 0.1 mm/s). Go 0.1 mm deeper from desired DV position to help create space for injection and move up before injecting.
- Inject virus by pressing RUN on the syringe controller.
- Leave needle in brain for 5 min after injection is complete.
- Avoid hitting the table during this time – needle will move inside brain.
- Consider preparing green surgery card or diluting retrobeads (if needed) while you wait.
- Raise the needle 0.1 mm. Wait 5-10 s. Repeat. This minimizes needle vacuum and dorsal spread of virus.
- Slowly rise needle from the brain (min 0.1 mm/s).
- When needle is far from skull, eject a bit of virus to clean the tip. Absorb bubble with a triangle sponge.
- Remove the syringe from the stereotax arm. Repeat these steps for all other injections.
- If injecting retrobeads, check out **Appendix 4 – Retrobeads**. Importantly, retrobeads can dehydrate and clog the needle. To keep this from happening, soak a triangle sponge with saline and gently stab the sponge with the needle. Position the syringe in a way that the tip of the needle is kept moist.

Surgery IV – Sutures (not used if doing implants)



(image only shows two loops)

- We use 2-3 knots with 3 loops each.
- Apply a drop of saline to the skull using the scrapper to re-hydrate the skin. Remove excess saline with sponge.
- Relax the ear bars a bit to allow the skin to move more freely.
- Left hand hemostat: gently lift the skin.
- Right hand hemostat: pull needle/suture through skin.
- Left hemostat: grab needle from the right hemostat and pull until a small tail is left (3-4 cm). Let go.
- Left hemostat: grab long end of the suture 5-10 cm away from incision.
- Right hemostat: move around the suture 2-3x clockwise. This will create a loop around the hemostat. Grab tail of suture and pull through the loop to make the first part of the knot. Let go.
- Left hemostat: adjust position if needed.
- Right hemostat: move around the suture 1-2x counterclockwise. Grab tail and pull. You made a square knot! Let go.
- Right hemostat: move around the suture 1-2x clockwise. Grab tail and pull. You finished the surgeon's knot. Let go.
- Cut suture ends – leave ~0.3 cm. Repeat these steps for the next 1-2 knots.
- Release ear bars, push iso mask away and remove mouse from bite bar.
- Gently hold the mouse head up and apply Vetbond (blue glue) to the sutured skin. Keep the head up for a bit to avoid glue from reaching eyes.
- Ear tag if needed. Gently put mouse in recovery bin (should be heated).
- Turn OFF flow switches for iso mask on anesthesia machine.

Surgery V – Implants

- Keep porcelain dish on ice or inside the fridge.
- To help Metabond bind well, score the skull more if needed and dry it well – use alcohol wipes and swab/sponge.
- Remove any blood clots or debris from drill holes using forceps.
- Grab a cannula holder – it should have a white ferrule holder at the tip. If not (like figure below), give it one.



- Insert probe into holder – insert and remove the probe a few times to loosen its grip a bit.
- Make sure probe is vertical and not tilted.
- Attach cannula holder to stereotax arm – note that we have holders for the left and the right arm.
- If doing 2 or more implants, set up a second cannula holder and attach it to the other stereotax arm.
- Zero probes at bregma. Implant the probe further away from bregma first.
- If probes will be very close, consider using eye ointment to block the spread of metabond into one of the skull holes.
- Move probe to target coordinates and lower it into brain slowly – DV position should be 0.1 mm above virus injection.
- Use a triangle sponge to absorb any blood and keep the skull dry.
- In the cold porcelain dish, mix: 1 scoop of metabond powder, 2 drops of metabond liquid and 1 drop of catalyst.
- Mix with green applicator and apply to the exposed fiber optic and ferrule. Let metabond drip/run to the skull. Fiber optic should be completely covered by metabond.
 - Do NOT apply metabond to the ferrule holder or the cannula holder. If you do, quickly wipe it off with skull scrapper and/or triangle sponge.
 - AVOID applying metabond to the mouse's skin.
 - Use a new green applicator for every probe implant.
- If only doing 1 implant, generously cover the skull with metabond. If doing more implants, avoid applying metabond to bregma or other skull holes.
- Wait ~5 min for metabond to dry. During this time, clean the porcelain dish with a kimwipe. Do NOT let metabond dry on the porcelain dish – it is a struggle to clean it afterwards.
- Check if metabond is dry with skull scrapper.
- Hold ferrule with forceps and gently rise the cannula holder. If you accidentally applied metabond to the ferrule holder or if you did not wait long enough, you will know now: you will pull the probe out. SADNESS.
- Repeat the steps above for all other implants.
- Test the Flow-It ALC Light Curing Cement on your sterile pad (you might have to push the plunger a lot before it starts running, but it will leak continuously after that).
- Apply the cement around and between the probes. Leave 5 mm of each ferrule exposed for connection with the optic cords later. Try to make a smooth hill covering all metabond. Pull the plunger to stop the cement from leaking out.
- Cure cement with UV light (should be charging by stax 2). Do NOT look directly into the light source – shield eyes with red cover. Press the power button a few times until it turns ON. Move the UV light around the cement hill. When light turns OFF, turn it ON again.
- Check smoothness of the cement hill with the skull scrapper.
- (Optional) Apply Vetbond around the cement to seal skin.
- Ear tag if needed. Remove mouse from stereotax and place it in the recovery bin (should be heated).
- Turn OFF flow switches for iso mask on anesthesia machine.



This mouse will get 3 probes! One probe is implanted, one is already mounted (on the left), but I need bregma visible to zero the third probe. Note that metabond is covering the optic fiber completely. This amount of metabond was enough to keep the probe in place when I removed the cannula holder.

After surgery – Mouse recovery

- Put wet food (DietGel) inside cage. Also move some dry food to the cage floor.
 - If we do not have DietGel, moisten some of the dry food with water.
- Fill out green card with surgery info and put it in front of the cage card.
- Monitor mouse's recovery – mice are ready to return to home cage when awake and mobile. A good indicator of anesthesia recovery is grooming – mice will often clean up the eye ointment. Longer surgeries will need a longer recovery period.
- Check on animal 24 h later and report status on green card. If animal shows any signs of lethargy, bad coat condition, or open mouth breathing, consider euthanasia. We do not remove stitches.

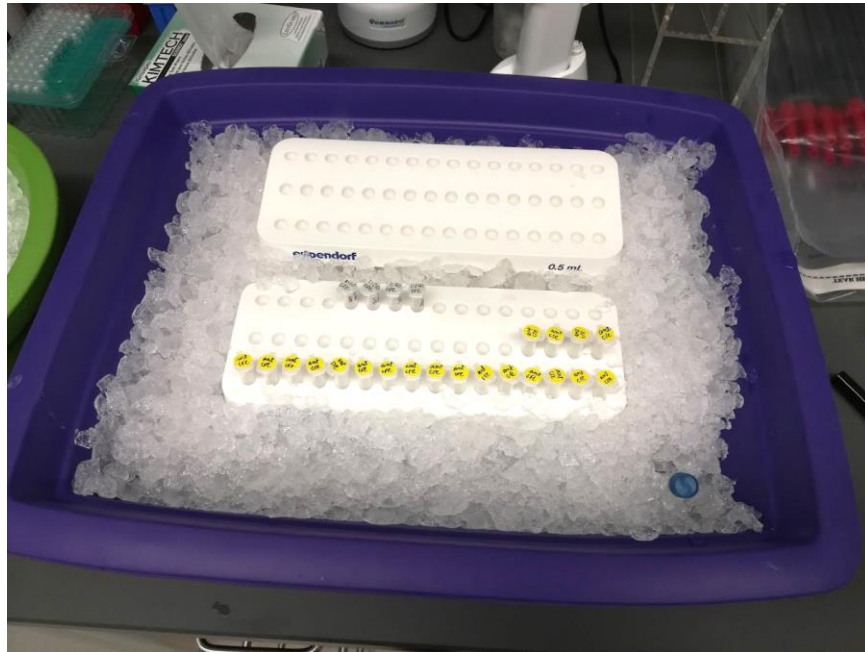
After surgery – Clean-up

- Collect any leftover virus from the syringes you wish to store. Otherwise, use continuous infuse to empty syringes.
- Clean NanoFil syringe(s) with saline and then water. Store them carefully.
- Label and store leftover carprofen in the fridge.
- Discard used scalpel blade and leftover suture in sharps container.
- Discard all surgery waste in the biohazard bin.
- Clean metal tools, ear bars, bite bar, drill, and heated pad cover with alcohol wipes.
- Do a Full Evac of anesthesia machine.
- Turn OFF all equipment:
 - Lights
 - Drill
 - Digital stereotaxic display
 - Syringe pump controller
 - Heating pad – plug battery back in to recharge
 - Recovery heating pad
 - Germinator (bead sterilizer)
 - All flow switches in anesthesia machine
 - Iso Vaporizer in anesthesia machine
 - System pressure switch in anesthesia machine
 - ON/OFF switch in anesthesia machine
- Close oxygen tank.
- Return all cages to their rooms.
- Dispose of solid waste in the recovery bin and induction chamber.
- Clean recovery bin and induction chamber with CDOX sprayed paper towels.
- Dispose of full charcoal canister in biohazard waste.
- Leave empty iso bottles open inside the hood. After airing out, empty bottles go into glass waste. (CHECK)
- Make a note of any supplies that need to be re-stocked. Inform our lab tech or someone who can order more supplies. If we have the supplies in lab, leave a note on the slack channel for the next person doing surgeries to take these supplies up.
- Place Bose speaker on its charging platform.
- Store ice bucket & leftover buprenorphine in the histology room.

Appendix 1 – Aliquoting viruses

Viruses should only be thawed twice – once for aliquoting and once for surgery – because repeated freeze-thaw cycles will damage virus particles and decrease the effective titer. Have an ice bucket ready when removing viruses from the -80. Try NOT to touch the bottom of tubes storing viruses, to avoid heating the virus with your fingers.

- Grab ice with the big purple bucket.
- Remove virus stock from -80 and leave it thawing slowly on ice.
- Put a tube rack on the ice (white plastic racks are in the histology room).
- Put small eppendorf tubes (0.5 ml) in the rack.
- The goal here is to keep the tubes cold during aliquoting to preserve the virus as much as possible.

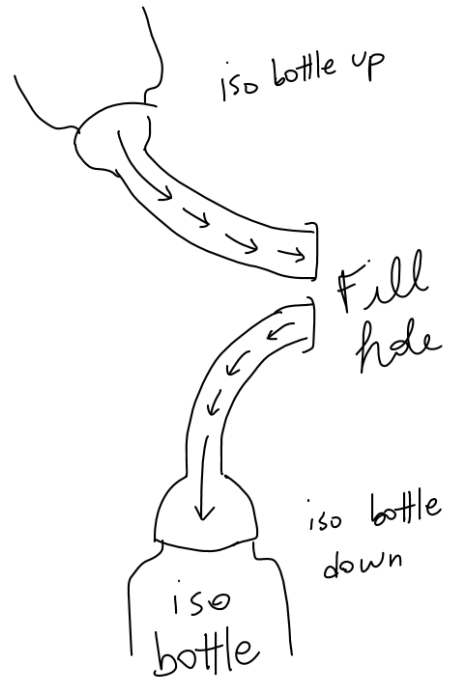
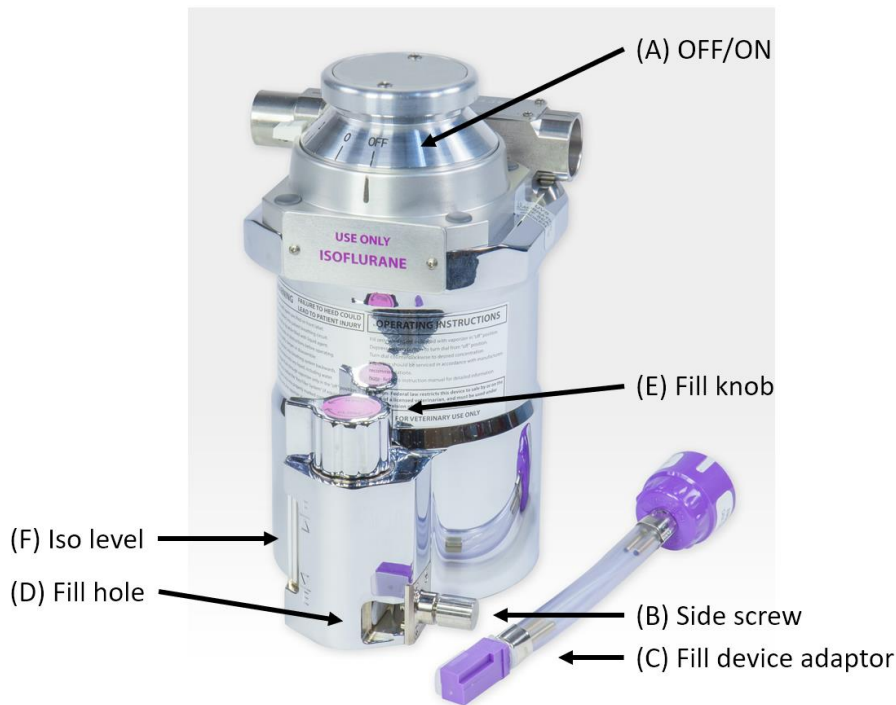


- Once virus is completely thawed (this can take ~30 min), put on your gloves.
- Do a quick spin of virus stock in a bench centrifuge for ~2 s.
- Make 5 and/or 10 ul aliquots – make sure to pipette stock up and down for a bit before aliquoting.
- Do NOT vortex the stock.
- Change pipette tip often if you notice leftover solution in tip.
- Discard pipette tips into the biohazard waste.
- Label tubes with colorful round labels – if you make aliquots of different volumes, choose a color for each volume.
- Label storage box with new titer and date.
- Update virus spreadsheet on the lab Google Drive.

Appendix 2 – Filling iso tank/vaporizer

Iso **SHOULD NOT** leak during this process. Follow the steps carefully to avoid leaks. If you accidentally spill iso, clean the spill with paper towels and put them inside the hood. Turn ON air flow in the hood to minimize inhalation. Consider stepping out of the room if spill was significant – protect yo'self!

- Make sure machine is OFF.
- Make sure vaporizer is OFF (A).
- Open side screw (B).
- Place fill device adaptor (C) on iso bottle and open its plastic cap.
- Keep fill device adaptor facing up and place it all the way into the fill hole (D). The open side of the adaptor should face away from the side screw.
- Close side screw tightly (B).
- **Move iso bottle up** until fill device adaptor fills up with iso.
- Turn fill knob (E) to OPEN and wait until iso reaches max level (F).
- Turn fill knob (E) to CLOSE.
- **Move iso bottle down** to empty the fill device adaptor.
- Open side screw (B) and remove adaptor from fill hole
- Close the plastic cap of the adaptor and store bottle with adaptor in the hood.



Appendix 3 – Changing gas regulator

- TO DO.
- Remove strips from tank label to mark it as “empty” or “in use”.
- Order new tank.

Appendix 4 – Retrobeads

- TO DO.